

YOUNGSEO SON

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Research Interests

My key research focus is in the field of Natural Language Processing (NLP) for social media analysis, language modeling, information extraction and data analysis. I collaborated with psychologists and computational linguists for user-based language modeling to obtain higher accuracies of various NLP tasks such as sentimental analysis, political stance detection, or sarcasm detection. I especially adapt discourse relation parsing to social media to detect which text spans contain key information for targeted tasks such as opinions or reasons for sentiment of reviews and a political stance.

Education

STONY BROOK UNIVERSITY , STONY BROOK, NY	AUG 2015 – PRESENT
Doctor of Philosophy in Computer Science, GPA – 3.89/4.00	
STONY BROOK UNIVERSITY , STONY BROOK, NY	AUG 2013 – MAY 2015
Summa Cum Laude, Honors Program B.S Computer Science, GPA – 3.93/4.00	
AJOU UNIVERSITY , SUWON, SOUTH KOREA	MAR 2009 – AUG 2013
B.E Computer Engineering, GPA – 4.36/4.50 (Major GPA – 4.43/4.50)	

Publication

Recognizing Counterfactual Thinking in Social Media Texts. **Youngseo Son**, Anneke Buffone, Anthony Janocko, Allegra Larche, Joseph Raso, Kevin Zembroski, H Andrew Schwartz, Lyle Ungar. In *ACL 2017*

Human Centered NLP with User-Factor Adaptation. Veronica E. Lynn, **Youngseo Son**, Vivek Kulkarni, Niranjan Balasubramanian, H. Andrew Schwartz. In *EMNLP 2017*

Capturing the Human Nuances of Optimism using Discourse Relations and Explanatory Style Recognition. **Youngseo Son**, Nipun Bayas, H Andrew Schwartz. In *MASC 2017*

Health Prediction based on Optimistic Explanatory Styles in Social Media. **Youngseo Son**, Laura Smith, H Andrew Schwartz. In *MASC 2016*

Predicting Users' Features from Their Relationships with Other Users. **Youngseo Son**, Yanhong Liu. In *Stony Brook URECA 2015*

Research Projects

The Language of Well-Being Project	JUL 2017 – PRESENT
(Project in collaboration with the University of California, Davis and the University of Melbourne)	
· Correlating linguistic features of people's everyday language with the changes of their emotions	
· Conducting LDA topic clustering over the transcripts of the participants' daily speech for the emotion analysis	
· Using N-gram, Linguistic Inquiry and Word Count (LIWC), sentiment/emotion lexicons	
9/11 World Trade Center Project	AUG 2017 – PRESENT
(Project in collaboration with Stony Brook WTC Wellness Program and Stony Brook Medicine)	
· Analyzing interviews of people who were at the scene of 9/11 WTC Attack	
· Correlating linguistic features of the subjects with their mental/physical health	
· Using LDA topic clustering, discourse relation parsing, sentiment/emotion lexicons	
Personality Analysis Using Linguistic Styles of Social Media Languages	MAY 2016 – PRESENT
(Project in collaboration with the University of Pennsylvania)	

- Leading the project with help of Prof. Ungar and and Anneke Buffone of the World Well-Being Project
- Analyzing correlations between syntactic structures of Facebook Status updates and profile information, personality, socio-economic statistics of users
- Worked as a Summer Intern Researcher in the Positive Psychology Center of the University of Pennsylvania

Health Prediction based on the Explanatory Style Analysis of Social Media

AUG 2015 – PRESENT

(Project in collaboration with the University of Pennsylvania and Prof. Schwartz)

- Presented at Mid-Atlantic Student Colloquium on Speech, Language and Learning (MASC-SLL 2016 and 2017)
- Using state-of-the-art discourse dependency parsers such as RST parsers or PDTB parsers for distracting discourse relations from messages collected from Facebook
- Analyzing Facebook status updates in order to predict users' personality and mental health.
- Data collection using Amazon Mechanical Turk

Privacy Policy Language Modeling using Prolog

AUG 2014 - May 2015

(Senior Honors Research Project under Advisor Professor Y. Annie Liu & Logic Programming Group of Stony Brook)

- Facebook privacy policy translation for Object-Oriented Modeling and XSB, Flora for the Relational Modeling
- Creation of automatic policy conflict detector aiming to prevent discrepancies between Facebook privacy policy and the terms of use of Third-Party partners

User Profile Prediction Based on Relationships in Social Networks

SUMMER 2014

(Undergraduate Research & Creative Activities (URECA) 2014, Stony Brook University)

- Users profile information prediction from their relationships with other users in Google+, Facebook, and Twitter dataset from Stanford Network Analysis Project
- Outlier Detection to social network relationships with Local Outline Score from "Oddball"
- Published in Stony Brook URECA Collected Abstracts and made a presentation for the Celebration of URECA Symposium

Work Experiences

Stony Brook University Research Assistant

SUMMER 2017 - PRESENT

- Funded by Stony Brook Medicine
- Developed a docx parser for the analysis using Differential Language Analysis Toolkit (DLTK)
- Correlating languages with human variables (e.g., personality, physical/mental health, income) by using language models including discourse relation parsing and topic clustering

World Well-Being Project (WWBP), the University of Pennsylvania, Research Scientist Intern

SUMMER 2016

- Developed a discourse relation parser for social media to capture counterfactual thinking from tweets
- Led the project with the help of Prof. Lyle Ungar and Anneke Buffone of the WWBP team

Dassault Systemes Korea Intern Software Engineer Internship

DEC 2012 - FEB 2013

- Development of Product Lifecycle Management (PLM) Web Application (**ENOVIA**)
- Worked on updating PLM chart display and data visualization functions with PLM Development senior software engineers and sales team members

Korean Augmentation to the US Army (Korean Army Sergeant)

AUG 2010 - MAY 2012

- Worked as an Information Technology Specialist (25B)
- Worked with 2nd Infantry Division 8th Army of the United States deployed to South Korea
- Managed and did troubleshooting on networks, computers, peripheral devices, and online portal of the battalion

Teaching Experience / Activities

Stony Brook University Computer Science TA

SPRING 2014 – PRESENT

- Teaching Assistance for Big Data Analytics, Senior Software Engineering Project, Computer Science III (Java)

Programming), Advanced Game Programming, and Computer Music	
Elementary School Science Lab Teacher (Graduate Student Volunteers)	SPRING 2016 – PRESENT
· Give special topic science labs to the fifth graders of the Nathaniel Woodhull Elementary School	
Stony Brook University Korean Graduate Students Association Vice President	FALL 2017
Korean-American Scientist and Engineers Association (KSEA) Volunteer	FALL 2016, FALL 2017
Ajou-Stony Brook University Students Association President	SPRING 2015
Stony Brook Computing Society	FALL 2013 – PRESENT
· Participated in Hackathon competitions	
· Participated in computer science related presentation and seminars	

Skills

- Java, Python, MATLAB, C, C++, MFC
- Web Development (Javascript, JSP, XML, HTML, CSS, MySQL, Apache Tomcat)
- Versioning (SVN, GIT)
- Linux systems (Ubuntu 14.04), Mac OSX, and Windows
- Shell Scripts (Bash, Windows batch)
- Documentation (Latex, MS Offices)

Awards / Scholarship

Special CS Department Chair Fellowship	AUG 2015
Stony Brook University Computer Science Award of Honor	MAY 2015
Stony Brook University URECA Stipend	SUMMER 2014
Stony Brook University Outstanding Academic Achievement Awards	FALL 2013 - SPRING 2014
Stony Brook University Dean's List Nomination	FALL 2013 - SPRING 2014
Ajou University Superior Academic Performance Scholarship	FALL 2009 - SPRING 2013
US Army Best Korean Augmentation to the United States Army (KATUSA) of 2012	MAY 2012
US Army Best Warrior 2012	APR 2012